

Supplementary Materials

An artificial intelligence modeling framework based on microbial community structure prediction enhances the pollutant removal efficiency of the algae-bacteria granular sludge system

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Text S1. The characteristics of the reactor

Photo-sequencing Batch Reactor (PSBR) had a height of 1100 mm, an inner diameter of 70 mm, resulting in a height-to-diameter ratio (H/D) of 15.7, and an effective volume of 3 L. During operation, the reactor was placed in a shaded location with LED lights set up around it to provide a constant intensity of illumination. The influent water for the experiment entered through the top of the reactor via a water pump, with the liquid level controlled by a liquid level relay. The drain outlet was located in the middle of the reactor and controlled by a solenoid valve, with a water exchange volume of 50% per cycle, amounting to 1.5 L of water exchanged. A sand-core aeration head was installed at the bottom of the reactor for aeration, with the aeration rate controlled at 2 L/min by a glass rotor flowmeter. The PSBR reactor operated at room temperature with a cycle duration of 6 h, and the operating time of each phase was regulated by a time controller, including five stages: influent feeding, aeration, sedimentation, drainage, and idle. Among these, both influent feeding and drainage lasted for 5 min, followed by 30 min of anaerobic idling after influent feeding, a sedimentation time of 5 min, and an aeration time of 315 min. In this study, LED lights were used as the light source and placed at the rear side of the reactor, with the light intensity set to 5000 ± 200 lux close to the inner wall of the reactor. The light/dark cycle was set at 12h light/12h dark (Bao et al., 2024). The reactor was operated with synthetic wastewater. The carbon source, nitrogen source, and phosphorus source were sodium acetate, ammonium sulfate, and potassium dihydrogen phosphate, respectively (Liu et al., 2024). The inoculated sludge was algal-bacterial granular sludge (ABGS) previously cultivated by our research group.

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Text S2. Extraction and determination of chlorophyll

The experiment characterized the growth status of microalgae based on chlorophyll content. Chlorophyll extraction was conducted using the hot ethanol extraction method (Bao et al., 2024), and the chlorophyll content was calculated using the following formula:

$$\text{Chla}=0.0604A_{632}-4.5224A_{649}+13.2969A_{665}-1.7453A_{696} \quad (\text{S1})$$

$$\text{Chlb}=-4.1982A_{632}+25.7205A_{649}-7.4096A_{665}-2.7418A_{696} \quad (\text{S2})$$

$$\text{Total Chl}=24.1209A_{632}+11.2884A_{649}+3.7620A_{665}+5.8338A_{696} \quad (\text{S3})$$

Text S3. High-throughput analytical methods for microbial communities

Total genomic DNA was extracted from sludge samples using the Ezup Column Soil DNA Extraction Kit from Sangon Biotech (Shanghai) Co., Ltd. The concentration and purity of the DNA were then checked using a Nanodrop2000 spectrophotometer, and the DNA samples that met the criteria were stored at -20°C. Sequencing was performed by Shanghai Personal Biotechnology Co.,ltd. (Shanghai, China) on the MiSeq platform (Illumina, USA). The upstream primer for bacterial sequencing was 338F (5'-ACTCCTACGGGAGGCAGCAG-3'), and the downstream primer was 806R (5'-GGACTACHVGGGTWTCTAAT-3'). For algal sequencing, the upstream primer was 3NDF (5'-GGCAAGTCTGGTGCCAG-3'), and the downstream primer was V4-euk-R2R (5'-ACGGTATCTRATCRTCCTTCG-3') ([Liu and Salles, 2024](#)).

Text S4. Pearson correlation analysis

Pearson Correlation Coefficient (PCC) analysis is a frequently employed statistical method used to assess the strength and direction of a linear relationship between two variables (Lee, 2019). Its value ranges from -1 to 1, where -1 indicates a perfect negative correlation, 1 signifies a perfect positive correlation, and 0 suggests no correlation. The Pearson correlation coefficient is defined as the ratio of the covariance between two variables to their standard deviations. By estimating the covariance and standard deviations of the sample, the Pearson correlation coefficient r can be obtained (Edelmann et al., 2021).

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y} \quad (\text{S4})$$

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (\text{S5})$$

Where: X_i and Y_i represent the observed values of the variables X and Y, respectively; $\bar{X}$ and $\bar{Y}$ are the mean values of the variables X and Y, respectively.

Text S5. Six Machine Learning-Based Approaches to Predictive Model Training

In constructing the prediction model for the pollutant removal efficiency of the ABGS system, six machine learning algorithms were adopted for training in this study (Liu et al., 2024). To adjust as many hyperparameters as possible within the limited time available and derive the optimal six machine learning models, Bayesian optimization was employed in this study (Dao et al., 2024), iterating 30 times to identify the best hyperparameters (Ullah et al., 2024).

1. Linear regression

The Efficient Linear Regression (ELR) model rapidly fits a linear relationship between independent variables and dependent variables through optimization methods such as the least squares method, and is widely used in predictive analysis across various fields (Pan et al., 2024). By conducting data preprocessing, feature selection, and algorithm optimization, the efficiency and accuracy of the model can be enhanced, ensuring high performance even in complex environments (Badawi et al., 2024).

In this study, Bayesian optimization was used to tune the parameters of the ELR model, including the regularization strength (Lambda) ranging from $2.381e-07$ to 2380.9524 , the learner (SVM, least squares), and the type of regularization (ridge, LASSO). This process aimed to minimize the Mean Squared Error (MSE) as the objective function (Gomez et al., 2024).

2. Regression tree

The Tree Regression (TR) model is a machine learning model based on a tree-like structure used to solve regression problems. It divides the feature space into different regions and uses a constant to predict the value of the target variable within each region (Swarnam et al., 2024).

In this study, Bayesian optimization was employed to tune the maximum leaf size hyperparameter of the RT model, ranging from 1 to 30. This process aimed to minimize the MSE as the objective function (Gomez et al., 2024).

3. Tree Embedding

The Tree Ensemble (TE) model is a powerful machine learning technique (Asif et al., 2024) that combines the predictive capabilities of multiple decision tree models to improve overall prediction accuracy and stability. Common tree ensemble models that can be used for regression problems include Bootstrap Aggregating (Asif et al., 2024) (Bagging) and Least Squares Boosting (Chen, X. et al., 2024) (LSBoost).

The parameters of the ensemble methods (Bag and LSBoost), maximum leaf size (ranging from 1 to 20), number of learners (from 10 to 500), number of predictor variables to be sampled (from 1 to 12), and learning rate (between 0.001 and 1) for the TE models were tuned through Bayesian optimization. This process aimed to minimize the MSE as the objective (Dao et al., 2024).

4. Support Vector Regression

Support Vector Regression (SVR) is a supervised learning algorithm that aims to find an optimal hyperplane (or a set of hyperplanes) to separate data samples of different classes.

In this study. The parameters of the SVR model, including the kernel function (Gaussian, linear, quadratic, and cubic), kernel scale (ranging from 0.001 to 1000), box constraint (from 0.001 to 1000), Epsilon (between 0.05 and 4975), and whether to standardize the data, are tuned through Bayesian optimization. The objective of this process is to minimize the MSE as target (Keshun et al., 2024).

5. Gaussian process regression

Gaussian Process Regression (GPR) is a non-parametric Bayesian regression method that is based on the idea of Gaussian processes, treating observed data as a realization of a stochastic process (Wen et al., 2024).

In this study, the parameters of the GPR model are tuned through Bayesian optimization. The parameters include the basis function (constant, zero, and linear), the kernel function (anisotropic exponential, Nonisotropic Matern 3/2, Nonisotropic Matern 5/2, anisotropic rational quadratic, anisotropic squared exponential, isotropic exponential, Isotropic Matern 3/2, Isotropic Matern 5/2, isotropic rational quadratic, and isotropic squared exponential), the kernel length scale (ranging from 0.001 to 1000), Sigma (between 0.0001 and 395), and whether to standardize the data. The objective of this tuning process is to minimize the MSE as target (Jin and Xu, 2024).

6. Neural network

The Neural Network (NN) model is a computational model inspired by the structure and function of the human brain. The deep feedforward neural network is a typical deep learning model, and the NN model used in this study employs this deep learning architecture. To address the issue of overfitting in NNs, this study applies L^2 regularization. L^2 regularization penalizes large weight values by adding the square of the weights to the objective function, thereby reducing the complexity of the model and mitigating the risk of overfitting (Khan and Maity, 2024).

In this study, the parameters of the NN model are tuned through Bayesian optimization. The parameters include the number of fully connected layers (ranging from 1 to 3), the activation function (ReLU, Tanh, Sigmoid, and None), the regularization strength (Lambda) (ranging from 2.3256×10^{-7} to 2325.5814), the size of each fully connected layer (from 1 to 300), and whether to standardize the data. The objective of this tuning process is to minimize the MSE as target (Gomez et al., 2024).

Text S6. Multitasking Neural Networks and Bayesian Optimization Scope

The multi-task neural network (MTNN) enhances learning efficiency and model generalization by simultaneously learning multiple related tasks, sharing feature representations or network parameters. It reduces the number of parameters, lowers the risk of overfitting, and leverages the correlations between tasks to improve overall performance. Multitask neural networks have been widely applied in various fields ([Yin et al., 2023](#)).

In this study, the parameters of the MTNN model are tuned through Bayesian optimization. The parameters include the number of fully connected layers (ranging from 1 to 15), the activation function (ReLU, Tanh, Sigmoid, and None), the regularization strength (Lambda) (ranging from 2.3256×10^{-7} to 2325.5814), the size of each fully connected layer (from 1 to 600), and whether to standardize the data. The objective of this tuning process is to minimize the MSE as target 10. By optimizing these parameters, the study aims to improve the learning efficiency and generalization ability of the MTNN model, leveraging the benefits of multitask learning ([Gomez et al., 2024](#)).

Text S7. Model performance metrics

The performance of each developed model was evaluated using four model performance metrics analysed as MSE, Coefficient of Determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The formulas for these metrics are as follows (Zhu et al., 2023).

Eq. (6-9)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2 \quad (S6)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2} \quad (S7)$$

$$R^2 = \frac{\sum_{i=1}^n (\tilde{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (S8)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \tilde{y}_i| \quad (S9)$$

Where y_i is the experimental true value; $\bar{y}$ is the mean value of the experimental data; $\tilde{y}_i$ is the fitted value of the prediction.

Text S8. Evaluating the determinism and stochasticity of bacterial and algal community assembly based on neutral community modeling

The ecological process of microbial community assembly constitutes the mechanism underlying the formation of microbial communities, encompassing both deterministic and stochastic processes ([Liu and Salles, 2024](#)). Deterministic processes involve both biotic and abiotic factors, namely interspecific interactions and environmental filtering. In contrast, stochastic processes regard all species as ecologically equivalent, encompassing random events such as birth, death, dispersion, extinction, and speciation ([Guo et al., 2023](#)). Consequently, deterministic processes can be interpreted as a form of adaptation of microorganisms to environmental factors or as habitat preferences. Stochastic processes, on the other hand, can be understood as a random balance between losses and gains among microorganisms. To ascertain whether deterministic or stochastic processes dominate the ecological construction of algal-bacterial systems in two types of ABGS, this study employed the Neutral Community Model for assessment ([Dini-Andreote et al., 2015](#)). The Neutral Community Model effectively describes the relationship between the occurrence frequency of populations and their relative abundances. Specifically, R^2 represents the goodness of fit; a higher R^2 indicates a greater influence of stochasticity on the community assembly process, whereas a lower R^2 suggests a stronger influence of determinism. The m value, obtained from the fitting equation of the neutral model, represents the migration rate of all OTU in each sample within the community. A smaller m value indicates greater restrictions on species dispersion ([Lean and Jones, 2023](#)).

Text S9. Bioinformatics analysis and molecular ecological correlation network construction

Ecological interactions within microbial communities, such as those involving resource or space exchange, have been a subject of interest in microbial ecology. The interactions among species are recognized as the driving force promoting the ecological functions of microbial communities. Due to their importance, species interaction networks have been used to describe community structure for over a century. Although networks were initially applied to the study of food webs, this concept has been extended to microbial ecology to uncover ecological interactions. Consequently, the interactions among microorganisms within a community are more likely to be reflected through molecular ecological correlation networks, which can be established using a set of mathematical correlation methods (Han et al., 2024). Recently, molecular ecological networks have been widely used to explore microbiomes in both natural and artificial environments, such as soil, sediments, bioreactors, and wastewater treatment plants (Jia et al., 2024).

In ABGS systems, highly diverse microorganisms thrive on organic compounds rich in carbon (C), nitrogen (N), sulfur (S), phosphorus (P), and various trace elements, forming a complex network of ecological interactions based on competition for resources and space (Pederzoli et al., 2024). Currently, a series of graphical methods have been developed to construct co-occurrence networks of complex microbial interactions among species in activated sludge and natural environments. Previous studies on correlation networks have defined multiple relationships among species using similarity matrices or sparse multivariate regression analysis, respectively. Typically, nodes and links in the network represent species and interactions, respectively. The interactions are defined by positive or negative correlations, and as an intrinsic property of correlation analysis, the network is usually undirected, indicating specific interactions among species, such as competition and symbiosis. The method adopted in this study for differential analysis within the constructed microbial community network is Linear Discriminant Analysis Effect Size (LEfSe) (Pederzoli et al., 2024).

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238 **Text S10.** Linear discriminant analysis Effect Size

239 LEfSe employs linear discriminant analysis to analyze species differences between different
240 groups. By setting different LDA Scores as screening criteria, we identified the differential species in
241 ABGS communities under different bacterial-algal ecological construction modes. A higher LDA
242 Score indicates greater species variability ([Jia et al., 2024](#)). Given the large number of significantly
243 different bacterial and algal populations, this study set an LDA Score >4.0 as the criterion for
244 screening bacterial community analysis, and an LDA Score >3.5 as the criterion for screening algal
245 community analysis.

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Text S11. SHapley Additive exPlanations

.. correlation analysis primarily assesses the relationships between features by calculating SHAP values and interaction values (Wang et al., 2024). The calculation of SHAP values is based on Shapley values, with the formula given as:

$$\phi_i(f) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (\text{S10})$$

Where: $\phi_i(f)$ is the SHAP value for feature i ; N is the set of all features; and S is a subset of the feature set.

The SHAP method not only provides local explanations (i.e., explanations for individual predictions) but also reveals the overall impact of features through the calculation of global feature importance analysis (Wang et al., 2024).

Text S12. Partial Dependence Plot analyzes the response relationship between input indicators and target indicators

Partial Dependence Plot (PDP) correlation analysis is a visualization tool used to understand the impact of a particular feature in a model on the prediction results. Through PDP, changes in the model output can be observed as a specific feature varies, thereby analyzing the relationship between that feature and the target variable (Divine et al., 2024). The formula for it is:

$$PDP(x_j) = \frac{1}{N} \sum_{i=1}^N f(x_1, \dots, x_{j-1}, x_j, x_{j+1}, \dots, x_p) \quad (S11)$$

Where: f represents the prediction function of the trained model; N denotes the number of samples; $x_1, \dots, x_p$ represent all the features; and x_j is the feature being analyzed.

Text S13. Non-dominated sorting genetic algorithms and hyperparameterization

Non-dominated Sorting Genetic Algorithm (NSGA) is an evolutionary algorithm widely applied to multi-objective optimization problems, aiming to simultaneously optimize multiple objective functions by simulating the natural selection process (Chen, Z. et al., 2024). Its core idea is to maintain population diversity and convergence through non-dominated sorting and crowding distance. Individuals in the population are sorted and divided into different ranks (Pareto fronts). The rank of each individual is determined by the number of individuals it dominates. Within each non-dominated layer, the crowding distance of each individual is calculated to assess the sparsity of the individuals (Hossain et al., 2022). The formula for calculating the crowding distance is as follows:

$$d(i) = \sum_{m=1}^M \left(f_m^{\max} - f_m^{\min} \right)^{-1} \left(f_m(i+1) - f_m(i-1) \right) \quad (\text{S12})$$

Where: f_m represents the value of the m th objective function, and $d(i)$ represents the crowding distance of individual i . By using the crowding distance, NSGA can simultaneously consider the non-domination and uniform distribution of individuals during the evolution process, thereby better maintaining the diversity of the population and aiding in finding a better approximate Pareto optimal solution set (Hossain et al., 2022).

In this study, the problem addressed is a three-objective programming problem. To solve this problem, the algorithm parameters set are as follows: selection operation uses tournament selection, coefficient for the best front individuals is 0.25, population size is 1000, maximum number of generations is 1000, stopping generation is 1000, and fitness function value tolerance is $1e-100$ (Hossain et al., 2022).

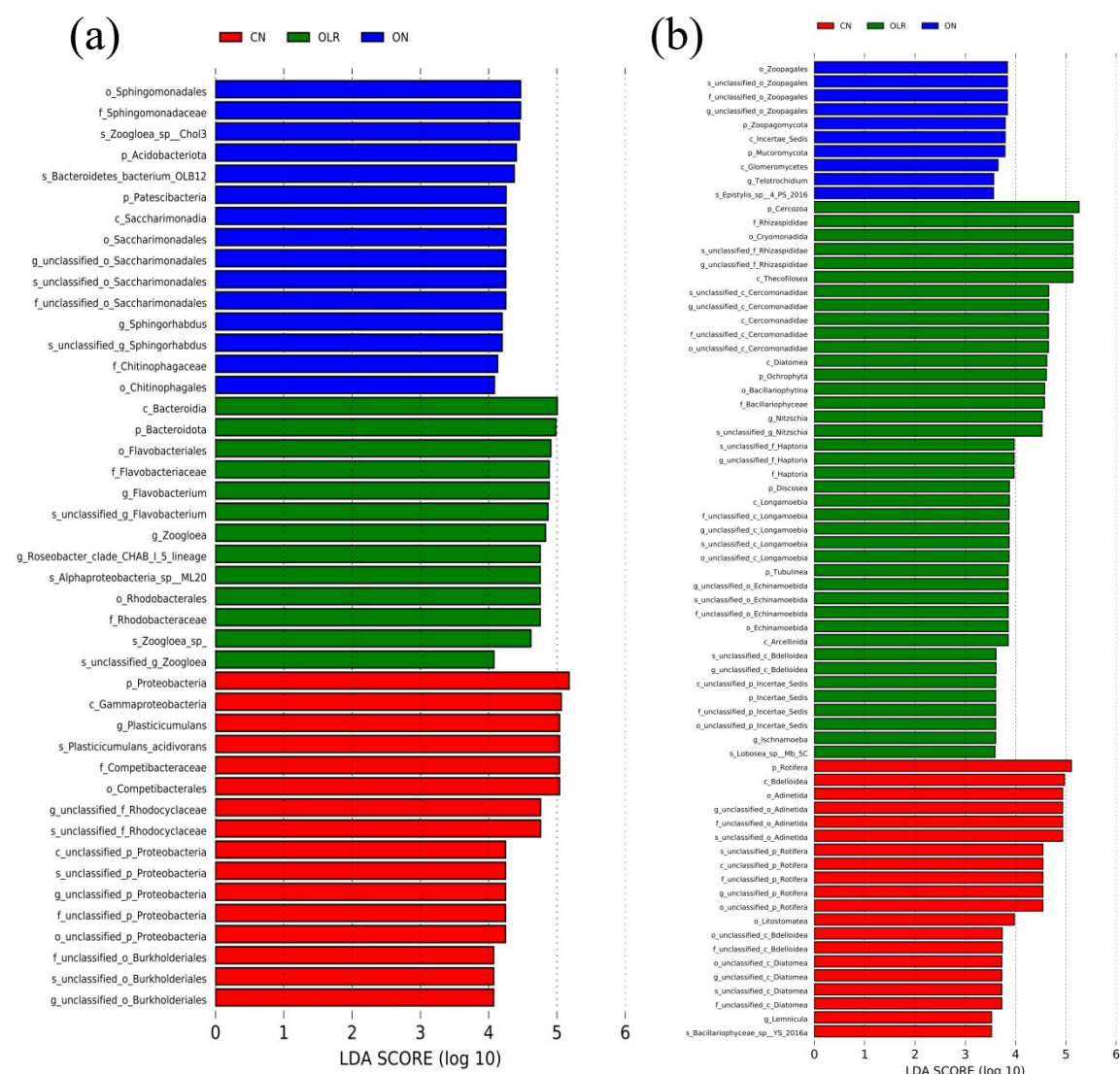


Fig. S1. Differential analysis of microbial community species under the regulation of different indicators. (a) Differential analysis of mycorrhizal species during microbial community succession under three kinds of regulation ($p < 0.05$), (b) Differential analysis of non-mycorrhizal species during microbial community succession under three kinds of regulation ($p < 0.05$).

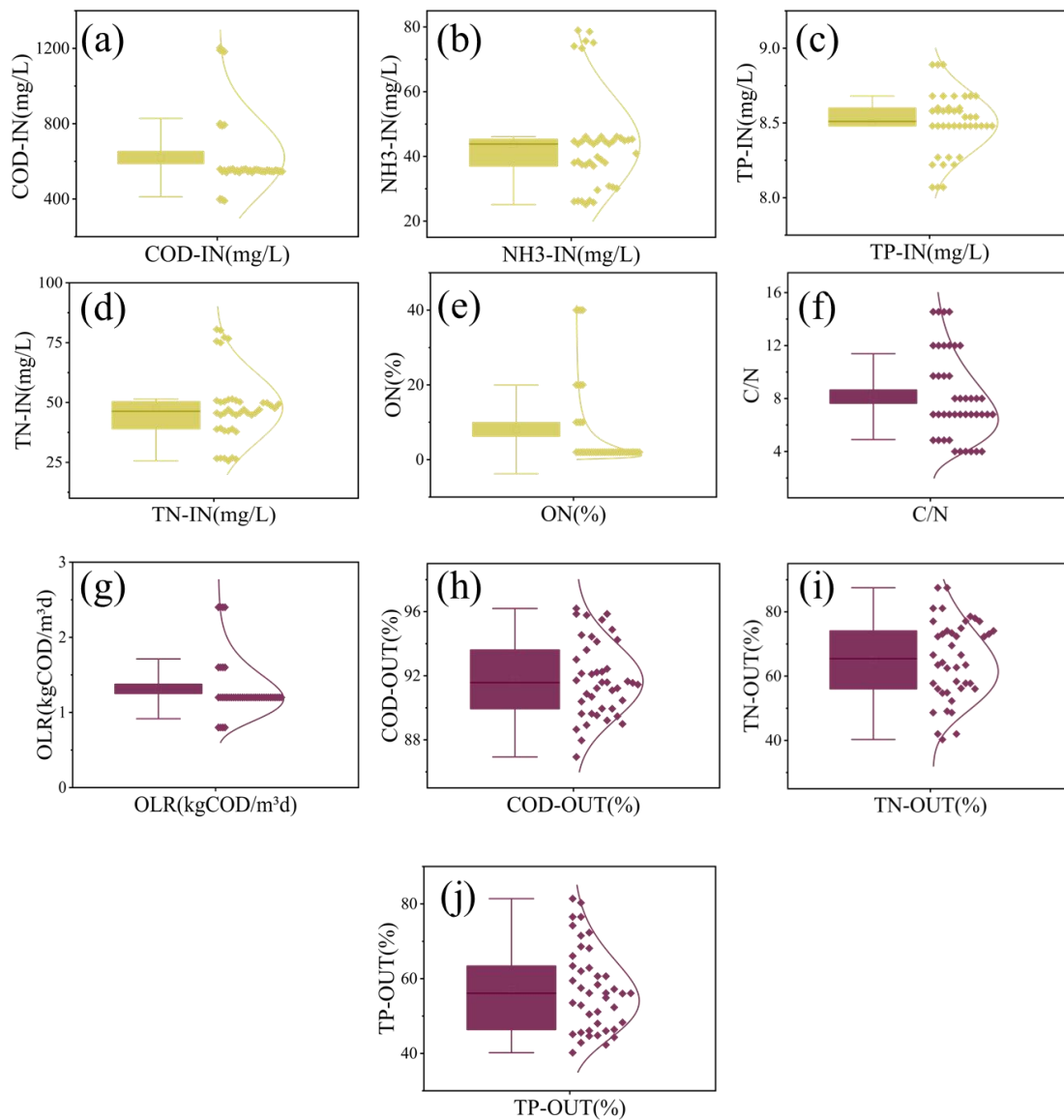


Fig. S2. Data Distribution and of Indicators in ABGS System. (a) COD-IN, (b) NH₃-IN, (c) TP-IN, (d) TN-IN, (e) ON, (f) C/N, (g) OLR, (h) COD-OUT, (i) TN-OUT, (j) TP-OUT.

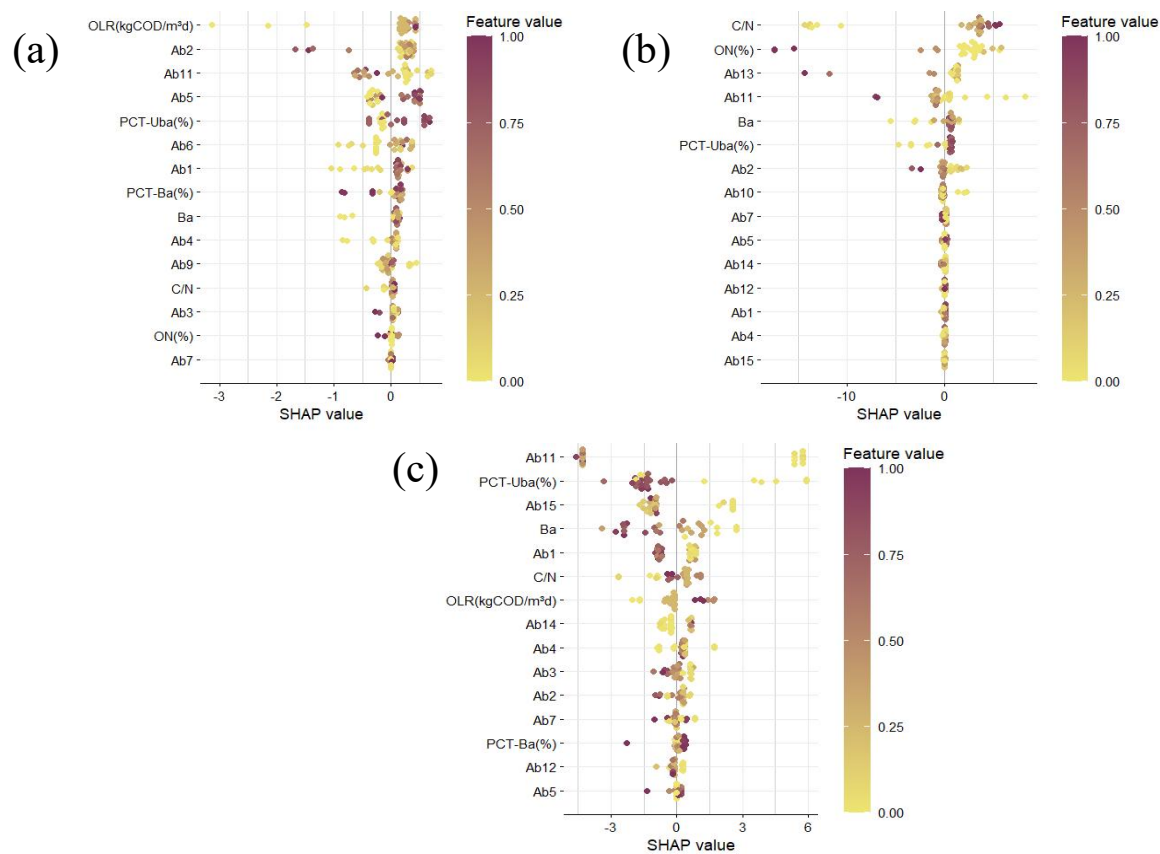


Fig. S3.(a) The SHAP importance analysis of the input index of the COD-OUT prediction model, (b) TN-OUT, (c) TP-OUT.

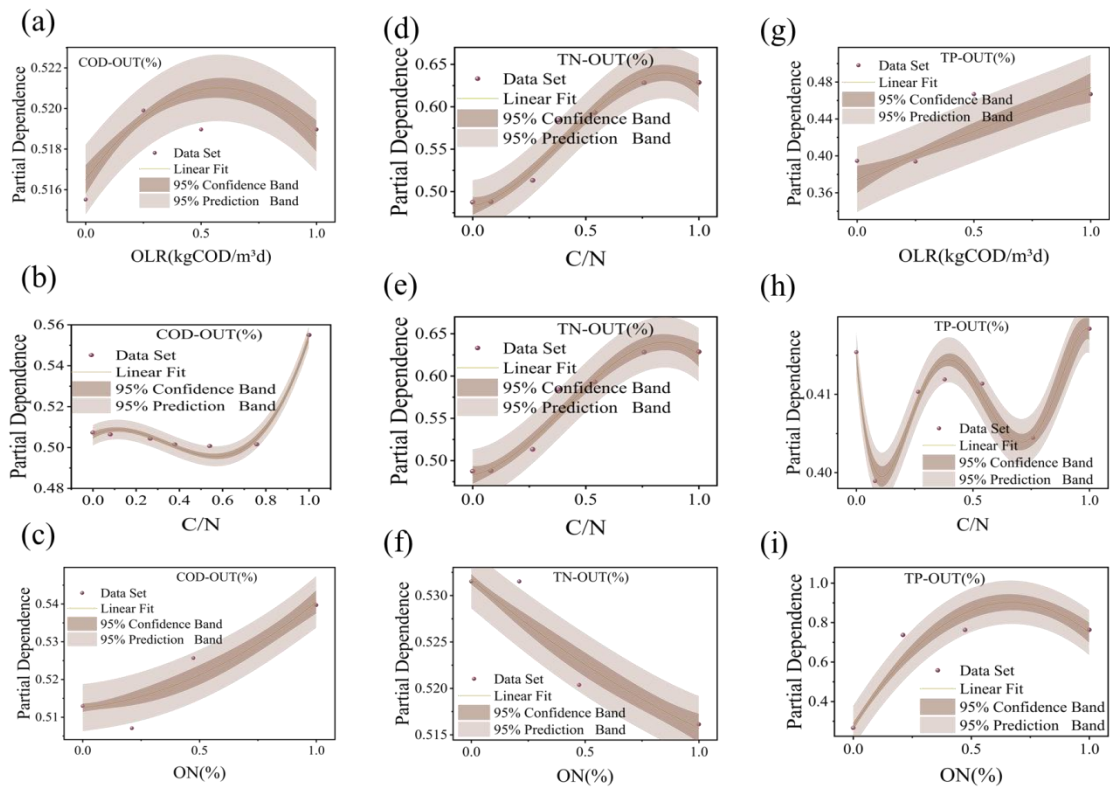
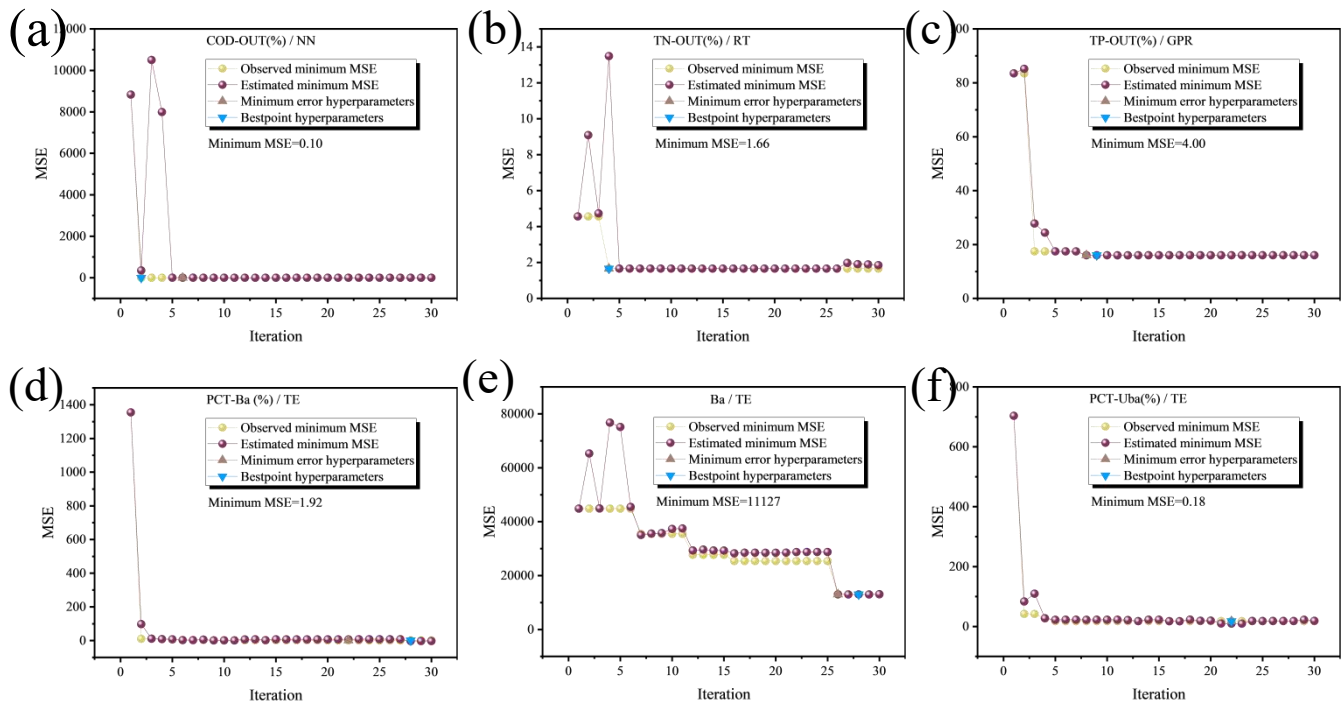


Fig. S4. (a) Plot of the mutual response of OLR to COD-OUT based on PDP, (b) Plot of the mutual response of the PDP-based regulatory index C/N to COD-OUT, (c) Plot of the mutual response of the regulatory index ON to COD-OUT based on PDP, (d-f) Analysis of the TN-OUT prediction model, (g-h) Analysis of the TP-OUT prediction model.



314 **Fig. S5.** The Bayesian optimization process for determining the optimal prediction models
 315 includes. (a) Bayesian hyperparameter optimization for the optimal machine learning model in
 316 predicting COD-OUT, (b) TN-OUT, (c) TP-OUT, (d) PCT-Ba, (e) Ba, (f) PCT-Uba.

Table S1

Descriptions of the features present in the dataset used for modeling.

Parameter	Description
Time (t) (day)	PSBR reactor incubation time for ABGS
C/N	BOD ₅ /TN in PSBR reactor feed water
OLR(kgCOD/m ³ d)	Organic loading rate of PSBR reactor
ON(%)	Organic Nitrogen in PSBR Reactor Feed Water
COD-IN(mg/L)	Chemical Oxygen Demand content in PSBR reactor feed water
TN-IN(mg/L)	Total nitrogen in PSBR reactor feed water
TP-IN(mg/L)	Total phosphorus in PSBR reactor feed water
NH ₃ -IN(mg/L)	Ammonia nitrogen content in the feed water of PSBR reactor
Ya(mg chl _a /SSg)*	Sludge chlorophyll a content in PSBR reactor
Yb(mg chl _b /SSg)*	Sludge chlorophyll b content in a PSBR reactor
MLSS(g/L)	Mixture liquid suspended solid in the feed water to the PSBR reactor
SVI ₃₀ (mL/g)	Sludge volume index of sludge in PSBR reactors
m(μm)	Granule size of ABGS in PSBR reactor
COD-OUT(%)	COD removal for one cycle in a PSBR reactor
TN-OUT(%)	TN removal for one cycle in a PSBR reactor
TP-OUT(%)	TP removal for one cycle in a PSBR reactor
PCT-Ba (%)	Percentage of bacterial species in the microbial community in the PSBR reactor
Ba	Number of bacterial species in the microbial community in the PSBR reactor
PCT-Uba(%)	Percentage abundance of dominant nonfungal species in microbial communities in PSBR reactors*
Ab	Abundance of dominant microorganisms in microbial communities in PSBR reactors*
Eq	The intervals of COD-OUT, TN-OUT, and TP-OUT values of the model's final regulation results indicate the equilibrium of carbon, nitrogen and phosphorus removal results
M	The microbial community structure index is composed of microbial abundance index (PCT-Uba and Ab) and diversity index (Ba and PCT-Ba)

*Chlorophyll was determined as described in Text S2, with dominance representing microbial species whose abundance ranked in the top 15 microbial species in microbial high-throughput sequencing.

Table S2

Range of input indicators for the two reactors

Parameter*	t(day)	C/N	OLR(kgCOD/ m ³ d)	ON(%)	COD-IN(mg /L)	TN-IN(mg /L)
max	30	14.55	2.40	40.00	1202.69	80.56
min	10	4.00	0.80	2.00	390.54	25.62
Parameter*	TP-IN(mg /L)	NH ₃ -IN(mg /L)	Y _a (mg chl _b /SSg)	Y _b (mg chl _b /SSg)	MLSS(g/L)	SVI ₃₀ (mL/ g)
max	8.89	78.95	2.97	0.46	4.51	92.00
min	8.07	25.11	1.38	0.14	2.50	40.16
Parameter*	m(μm)	COD-OUT(%)	TN-OUT(%)	TP-OUT(%)	PCT-ba (%)	ba
max	475.00	96.20	87.47	81.40	99.3	1537
min	319.20	86.93	40.28	40.18	87	161
Parameter*	PCT-Uba (%)	Ab				
max	65.93	51854.00				
min	1.19	0.00				

*The full name and interpretation of each acronym variable are given in Table S1.

Table S3

Bayesian optimized hyperparameters of optimal machine learning algorithms for 7 models

Model*	Optimal machine learning algorithms	Hyperparameters after Bayesian optimization
COD-OUT(%)	NN	Fully connected layers: 3
		First layer size:10
		Second layer size:10
		Third layer size:10
		Activation:None
TN-OUT(%)	RT	Regularization strength (Lambda): 3.75×10^{-7}
		Standardize data:Yes
		Minimum leaf size:1
		Basis function: Constant
		Kernel function: Nonisotropic Rational
TP-OUT(%)	GPR	QuadraticKernel
		scale:0.017885
		Sigma: 0.0001102
		Standardize data: Yes
		Ensemble method:LSBoost
PCT-ba (%)	TE	Minimum leaf size:5
		Number of learners: 24
		Learning rate:0.20076
		Number of predictors to sample:9
		Ensemble method:LSBoost
Ba	TE	Minimum leaf size:1
		Number of learners:41
		Learning rate:0.54871
		Number of predictors to sample: 5
		Ensemble method: LSBoost
PCT-Uba (%)	TE	Minimum leaf size. 4
		Number of learners: 11
		Learning rate:0.32983
		Number of predictors to sample:9
		Fully connected layers: 2
Ab	MTNN	First layer size:15
		Second layer size:17
		Activation:None
		Regularization strength (Lambda): 5.96×10^{-8}
		Standardize data:Yes

*The full name and interpretation of each acronym variable are given in Table S1.

Table S4

Performance metrics of six machine learning algorithms after Bayesian tuning of seven models

Model*	Machine learning algorithm	RMSE	MSE	R ²	MAE
COD-OUT(%)	TR	0.441	0.194	0.960	0.323
	ELR	2.113	4.465	0.078	1.916
	GPR	1.410	1.989	0.589	1.353
	TE	1.237	1.531	0.684	1.146
	NN	0.314	0.099	0.980	0.285
	SVR	0.347	0.120	0.975	0.279
TN-OUT(%)	TR	1.288	1.660	0.944	1.269
	ELR	4.241	17.984	0.397	4.186
	GPR	2.179	4.748	0.841	1.806
	TE	1.587	2.519	0.916	1.390
	NN	3.446	11.874	0.602	2.986
	SVR	2.988	8.926	0.701	2.969
TP-OUT(%)	TR	0.401	0.160	0.919	3.232
	ELR	13.373	178.840	0.093	9.731
	GPR	0.200	0.040	0.983	0.199
	TE	0.594	0.353	0.821	0.516
	NN	0.498	0.248	0.874	0.492
	SVR	0.461	0.212	0.892	0.448
PCT-Ba (%)	TR	1.867	3.487	0.903	1.399
	ELR	5.878	34.545	0.042	5.076
	GPR	1.628	2.650	0.926	1.376
	TE	1.387	1.923	0.947	1.146
	NN	5.618	31.564	0.125	3.750
	SVR	1.806	3.263	0.910	1.599
Ba	TR	170.933	29218.138	0.860	144.960
	ELR	253.603	64314.307	0.693	186.380
	GPR	149.877	22463.077	0.893	122.710
	TE	105.486	11127.215	0.947	80.223
	NN	139.648	19501.550	0.907	113.01
	SVR	151.440	22934.111	0.890	114.82
PCT-Uba(%)	TR	0.506	0.256	0.967	0.363
	ELR	0.739	0.546	0.930	0.492
	GPR	0.565	0.319	0.959	0.459
	TE	0.423	0.179	0.977	0.326
	NN	0.739	0.546	0.930	0.492
	SVR	0.620	0.385	0.951	0.423
Ab	MTNN	102.004	10405.005	0.951	98.445

*The full name and interpretation of each acronym variable are given in Table S1.

Table S5

Results of NSGA in regulating microbial communities under specific conditions

indicators*	ba	PCT-ba(%)	PCT-Uba(%)	Ab1	Ab2	Ab3	Ab4	Ab5	Ab6
Initial structure	1423	92.70	60.48	33111	16794	16554	975	9468	1326
NC	649	97.69	65.49	47465	61072	15092	21403	6277	27797
	649	97.69	67.09	47401	60130	14270	22205	6084	27730
	649	97.76	67.09	46568	63596	16622	20998	7263	26945
NP	1264	92.89	56.35	56944	50543	5367	17590	549	33503
	1264	92.89	58.56	57685	50570	5763	16461	899	34167
	1264	92.89	60.05	57541	50702	5429	16791	789	33786
NN	1260	92.77	55.37	54745	49886	9204	18197	10	35675
	1260	92.77	57.58	54724	49276	8653	18688	230	35817
	1260	92.77	57.25	54514	49731	9754	17770	273	36766
EQ	1260	92.89	57.20	51663	51754	6785	23149	2278	30476
	1260	92.89	59.41	53329	52318	7996	20405	1597	31838
	1246	92.89	59.41	49971	52823	6166	24861	3359	28517
indicators*	Ab7	Ab8	Ab9	Ab10	Ab11	Ab12	Ab13	Ab14	Ab15
Initial structure	14630	10501	3549	4482	2949	4981	28	1845	2009
NC	6334	11712	13464	12711	5829	1015	3930	5581	5987
	6444	11189	13378	12706	5637	984	3819	5576	6014
	6116	12855	13804	12764	6134	1034	4175	5746	5969
NP	9488	10225	13401	12631	4801	1980	3200	3855	5457
	9485	10471	13372	12547	4964	2041	3249	3746	5419
	9639	10623	13506	12621	4902	2059	3260	3773	5415
NN	7209	6329	11167	11951	5034	1223	2887	4111	5728
	7253	5778	10986	11893	4891	1189	2789	4085	5728
	6926	4747	10196	11696	5099	967	2783	4037	5746
EQ	8138	8366	12830	12681	4429	1309	3048	4708	5810
	8117	9181	12899	12581	4822	1472	3197	4491	5726
	8412	8878	13263	12947	4266	1268	3134	4939	5827

*The full name and interpretation of each acronym variable are given in Table S1.

351

352 **Table S6**

353 Top 15 species of the microbial community of the ABGS system under regulated OLR

Sort	Kingdom	Phylum	Class	Order	Family	Genus	Species
Ab1	Rhizaria	Cercozoa	Thecofilosea	uncultured			
Ab2	Bacteria	Proteobacteria	Alphaproteobacteria	Rhodobacterales	Rhodobacteraceae	Roseobacter_clade_CHAB-I-5_lineage	Alphaproteobacteria_sp._ML20
Ab3	Metazoa_Animalia	Rotifera	Bdelloidea	Adinetida			
Ab4	Bacteria	Bacteroidota	Bacteroidia	Flavobacteriales	Flavobacteriaceae	Flavobacterium	
Ab5	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	Zoogloea_sp.
Ab6	Stramenopiles	Ochrophyta	Diatomea	Bacillariophytina	Bacillariophyceae	Nitzschia	Nitzschia_amphibia
Ab7	Rhizaria	Cercozoa	Cercomonadida				
Ab8	Alveolata	Ciliophora	Intramacronucleata	Conthreop	Oligohymenophorea	Telotrochidium	uncultured_eukaryote
Ab9	Bacteria	Bacteroidota	Bacteroidia	Cytophagales	Microscillaceae	OLB12	Bacteroidetes_bacterium_OLB12
Ab10	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Dechloromonas	
Ab11	Bacteria	Proteobacteria	Gammaproteobacteria	Competibacteriales	Competibacteraceae	Plasticicumulans	Plasticicumulans_acidivorans
Ab12	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	Zoogloea_sp._Chol3
Ab13	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	
Ab14	Bacteria	Patescibacteria	Saccharimonadia	Saccharimonadales			
Ab15	Alveolata	Ciliophora	Intramacronucleata	Conthreop	Oligohymenophorea	Telotrochidium	

354

355

356 **Table S7**

357 Top 15 species of the microbial community of the ABGS system under regulated C/N

Sort	Kingdom	Phylum	Class	Order	Family	Genus	Species
Ab1	Rhizaria	Cercozoa	Thecofilosea	Cryomonadida	Rhizaspididae	Rhogostoma	
Ab2	Bacteria	Proteobacteria	Gammaproteobacteria	Competibacterales	Competibacteraceae	Plasticicumulans	Plasticicumulans_acidivorans
Ab3	Metazoa_Animalia	Rotifera	Bdelloidea	Adinetida			
Ab4	Bacteria	Proteobacteria	Alphaproteobacteria	Rhodobacterales	Rhodobacteraceae		
Ab5	Metazoa_Animalia	Rotifera	Bdelloidea	Adinetida			
Ab6	Alveolata	Ciliophora	Intramacronucleata	Conthreop	Oligohymenophorea		
Ab7	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae		
Ab8	Stramenopiles	Ochromyces	Diatomea	Bacillariophytina	Bacillariophyceae	Nitzschia	Nitzschia_amphibia
Ab9	Metazoa_Animalia	Rotifera	Bdelloidea				
Ab10	Bacteria	Proteobacteria					
Ab11	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Azoarcus	
Ab12	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Dechloromonas	
Ab13	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales			
Ab14	Bacteria	Bacteroidia	Bacteroidia	Cytophagales	Microscillaceae	OLB12	bacterium_TG141
Ab15	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	

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359

360 **Table S8**

361 Top 15 species of the microbial community of the ABGS system under regulated ON

Sort	Kingdom	Phylum	Class	Order	Family	Genus	Species
Ab1	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	
Ab2	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	
Ab3	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	
Ab4	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Dechloromonas	
Ab5	Bacteria	Proteobacteria	Gammaproteobacteria	Xanthomonadales	Xanthomonadaceae		
Ab6	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Azoarcus	
Ab7	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales			
Ab8	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	
Ab9	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Azoarcus	
Ab10	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	Zoogloea_ramigera
Ab11	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Comamonadaceae		
Ab12	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	Zoogloea_ramigera
Ab13	Bacteria	Proteobacteria	Gammaproteobacteria	Burkholderiales	Rhodocyclaceae	Zoogloea	Zoogloea_ramigera
Ab14	Bacteria	Proteobacteria	Gammaproteobacteria	Xanthomonadales	Rhodanobacteraceae	Aquimonas	Pseudomonas_sp._RL
Ab15	Stramenopiles	Ochrophyta	Diatomea	Bacillariophytina	Bacillariophyceae	Nitzschia	Nitzschia_amphibia

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363

364 **Table S9**

365 ABGS system running data for the first 1 month of 9 reactors

Reactor number	1	1	1	2	2	2	3	3	3	4	4	4	5	5
Ya(mg chlb/SSg)	2.52	1.38	1.47	2.60	1.99	1.90	2.59	2.61	2.60	2.52	2.40	2.55	2.60	2.67
Yb(mg chlb/SSg)	0.26	0.14	0.14	0.27	0.18	0.18	0.25	0.23	0.29	0.27	0.34	0.36	0.27	0.33
MLSS(g/ L)	2.90	4.30	4.51	2.70	3.90	4.30	2.80	2.99	3.45	2.70	3.50	3.20	2.90	3.00
SVI30(m L/g)	73.4	42.7	40.1	70.0	51.8	49.1	73.5	62.5	57.5	69.6	71.4	88.3	64.1	66.6
m(μm)	5	3	6	0	2	5	7	2	7	3	3	9	4	7
Time	381.	451.	475.	380.	429.	451.	382.	420.	443.	319.	322.	357.	338.	349.
	20	12	00	00	10	00	13	41	30	20	40	33	20	60
	1	11	31	1	11	31	1	11	31	1	11	31	1	11
C/N	14.5	14.5	14.5	9.70	9.70	9.70	4.85	4.85	4.85	4.00	4.00	4.00	8.00	8.00
	5	5	5											
OLR(kgC OD/m ³ d)	2.40	2.40	2.40	1.60	1.60	1.60	0.80	0.80	0.80	1.20	1.20	1.20	1.20	1.20
ON(%)	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
COD-IN(mg/L)	1202	1190	1189	793.	790.	793.	396.	400.	396.	558.	546.	551.	558.	546.
	.69	.00	.77	49	00	66	66	00	31	51	51	66	51	51
TN-IN(m g/L)	45.5	44.7	45.5	45.5	44.7	45.5	45.5	44.7	45.5	75.5	80.5	74.9	38.8	39.2
	9	4	9	9	4	9	9	4	9	8	6	4	5	4
TP-IN(mg /L)	8.58	8.68	8.89	8.58	8.68	8.89	8.58	8.68	8.89	8.48	8.68	8.22	8.48	8.68
NH3-IN(mg/L)	44.6	43.8	44.6	44.6	43.8	44.6	44.6	43.8	44.6	74.0	78.9	73.4	38.0	38.4
	8	4	8	8	4	8	8	4	8	7	5	4	7	5
Ba	646.	701.	966.	690.	716.	1038	671.	662.	1068	1211	1261	1423	1267	1537
	00	00	00	00	00	.00	00	00	.00	.00	.00	.00	.00	.00
PCT-Ba(%)	88.3	89.9	84.0	89.2	88.7	84.1	87.0	88.0	84.0	92.1	90.6	92.7	91.3	93.5
	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Ab1	4948	3078	1705	4700	4058	2265	3829	2722	2503	1341	2157	3311	4594	5185
	3.00	6.00	0.00	9.00	5.00	5.00	3.00	8.00	1.00	3.00	0.00	1.00	6.00	4.00
Ab2	1626	9042	1187	2302	9524	2163	2368	1520	1399	1958	4043	1679	2252	1017
	7.00	.00	1.00	6.00	.00	7.00	0.00	3.00	5.00	4.00	6.00	4.00	0.00	5.00
Ab3	9471	2471	2126	7945	1226	1178	1361	1548	2175	1520	1453	1655	8589	7070
	.00	0.00	3.00	.00	8.00	1.00	0.00	0.00	3.00	7.00	5.00	4.00	.00	.00
Ab4	9909	5068	1116	1682	1358	7837	1469	1854	2180	416.	292.	975.	8325	1013
	.00	.00	.00	7.00	3.00	.00	4.00	1.00	6.00	00	00	00	.00	6.00
Ab5	8980	1330	7890	7818	1387	1235	4973	8282	1067	1312	1061	9468	6876	6160
	.00	4.00	.00	.00	6.00	4.00	.00	.00	9.00	7.00	1.00	.00	.00	.00
Ab6	3102	1497	9561	3345	7217	8722	5636	9208	1151	2076	4800	1326	1121	2220
	.00	6.00	.00	.00	.00	.00	.00	.00	8.00	5.00	.00	.00	.00	.00
Ab7	8313	2409	6122	1368	1231	1378	7801	4856	3001	1037	1007	1463	8973	4566
	.00	.00	.00	8.00	2.00	1.00	.00	.00	.00	3.00	6.00	0.00	.00	.00
Ab8	6207	6131	1758	863.	1589	8574	648.	811.	5745	1224	1057	1050	3947	5003
	.00	.00	2.00	00	.00	.00	00	00	.00	.00	3.00	1.00	.00	.00
Ab9	3104	4101	1098	7384	8969	1338	4441	4418	452.	7621	4694	3549	3532	2574
	.00	.00	.00	.00	.00	.00	.00	.00	00	.00	.00	.00	.00	.00
Ab10	5498	3854	7397	3290	2647	4189	3563	2534	1820	852.	1992	4482	2533	5025
	.00	.00	.00	.00	.00	.00	.00	.00	.00	00	.00	.00	.00	.00
Ab11	604.	317.	95.0	4731	4110	2428	7325	6747	1315	2349	1801	2949	2288	4400
	00	00	0	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
Ab12	3548	5655	1251	2682	5020	1496	1276	3121	2002	1264	1237	4981	1780	2357
	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
Ab13	3214	1609	1046	4339	3117	1346	3010	3947	2877	8453	601.	28.0	5644	401.
	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	00	0	.00	00
Ab14	9231	5398	1305	1929	3483	1009	214.	163.	66.0	7074	4879	1845	366.	185.
	.00	.00	.00	.00	.00	.00	00	00	0	.00	.00	.00	00	00
Ab15	851.	1601	3714	831.	1461	2689	873.	829.	4943	4463	2153	2009	666.	1455
	00	.00	.00	00	.00	.00	00	00	.00	.00	.00	.00	00	.00
PCT-Uba(%)	56.2	62.5	69.4	50.5	53.9	55.9	51.4	48.1	56.6	56.5	51.2	60.4	56.8	65.9
	0	1	8	7	7	8	2	3	8	5	7	8	7	3
COD-OU T(%)	95.8	94.5	95.7	93.0	92.1	93.6	88.6	89.6	88.9	94.4	90.6	91.5	91.0	91.2
	6	4	8	1	4	1	5	3	2	4	9	9	9	2
TN-OUT(%)	77.0	81.0	87.4	72.3	73.0	74.0	48.6	42.0	56.0	40.2	52.2	56.2	57.7	73.4
	3	9	7	2	7	7	5	2	4	8	6	1	7	4
TP-OUT(%)	63.3	62.0	62.8	57.5	56.1	58.3	45.1	40.1	45.5	52.8	50.4	46.0	44.6	48.0
	8	0	8	0	2	7	1	8	6	8	7	7	5	0
Reactor number	5	6	6	6	7	7	7	8	8	8	9	9	9	
Ya(mg)	2.83	2.89	2.80	2.74	2.52	1.88	1.38	2.60	2.19	1.99	2.59	2.50	2.61	

chl _b /SS _g)													
Y _b (mg chl _b /SS _g)	0.27	0.24	0.30	0.30	0.26	0.20	0.16	0.27	0.18	0.20	0.25	0.33	0.25
MLSS(g/ L)	3.70	2.60	3.30	4.10	2.90	4.30	4.42	2.70	3.90	4.10	2.80	2.99	3.20
SVI ₃₀ (m L/g)	50.8 1	72.3 1	69.6 2	45.2 6	73.4 5	47.1 9	42.7 3	70.0 0	54.6 7	51.8 2	73.5 7	68.1 1	62.5 2
m(μm)	418. 00	343. 70	350. 90	426. 00	319. 20	366. 00	389. 00	338. 20	389. 00	408. 00	343. 70	391. 00	413. 00
Time	31	1	11	31	1	11	31	1	11	31	1	11	31
C/N	8.00	12.0 0	12.0 0	12.0 0	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80	6.80
OLR(kgC OD/m ³ d)	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20
ON(%)	2.00	2.00	2.00	2.00	10.0 0	10.0 0	10.0 0	20.0 0	20.0 0	20.0 0	40.0 0	40.0 0	40.0 0
COD-IN(mg/L)	551. 66	558. 51	546. 51	551. 66	552. 74	549. 17	547. 70	552. 74	549. 17	547. 70	552. 74	549. 17	547. 70
TN-IN(m g/L)	38.2 2	26.6 0	26.7 5	26.6 7	50.8 0	49.8 5	50.1 1	51.1 8	49.8 3	48.7 1	49.3 7	51.4 4	50.8 3
TP-IN(mg /L)	8.22	8.48	8.68	8.22	8.54	8.48	8.48	8.54	8.48	8.48	8.54	8.48	8.48
NH ₃ -IN(mg/L)	37.4 5	26.0 7	26.2 1	26.1 3	45.7 2	44.8 6	45.1 0	40.9 5	39.8 6	38.9 6	29.6 2	30.8 6	30.5 0
Ba	1514 .00	1223 .00	1366 .00	1619 .00	163. 00	161. 00	708. 00	181. 00	184. 00	731. 00	196. 00	163. 00	493. 00
PCT-Ba(%))	93.5 0	90.3 0	92.3 0	95.0 0	98.1 0	99.3 0	98.8 0	98.3 0	98.9 0	99.5 0	98.9 0	99.3 0	99.3 0
Ab1	1714 2.00	2618 7.00	3743 5.00	1108 9.00	2473 .00	2074 .00	272. 00	2486 .00	2196 .00	296. 00	3468 .00	3436 .00	1832 .00
Ab2	1179 3.00	3176 2.00	2797 7.00	5748 .00	2852 .00	2912 .00	68.0 0	3098 .00	2394 .00	65.0 0	2242 .00	2211 .00	2175 .00
Ab3	1264 0.00	1137 3.00	1047 8.00	1321 1.00	1702 .00	1696 .00	29.0 0	1704 .00	1350 .00	20.0 0	1355 .00	1383 .00	1108 .00
Ab4	2561 7.00	1357 8.00	1309 0.00	2678 3.00	343. 00	337. 00	9.00 00	612. 00	658. 00	0.00 00	580. 00	543. 00	711. 00
Ab5	1139 2.00	6862 .00	1395 2.00	1234 7.00	0.00 00	0.00 00	1889 .00	0.00 00	0.00 00	1813 .00	0.00 00	0.00 00	17.0 0
Ab6	6594 .00	1174 8.00	1575 .00	3086 8.00	469. 00	561. 00	0.00 00	369. 00	649. 00	42.0 0	524. 00	576. 00	477. 00
Ab7	1241 .00	1356 9.00	7293 .00	843. 00	583. 00	676. 00	0.00 00	437. 00	677. 00	10.0 0	395. 00	469. 00	305. 00
Ab8	753. 00	1142 5.00	7877 .00	910. 00	449. 00	415. 00	179. 00	315. 00	292. 00	161. 00	385. 00	320. 00	699. 00
Ab9	9699 .00	2383 .00	3043 .00	5262 .00	418. 00	573. 00	51.0 0	323. 00	482. 00	0.00 00	408. 00	456. 00	437. 00
Ab10	2400 .00	1448 .00	2875 .00	3707 .00	237. 00	100. 00	0.00 00	242. 00	135. 00	0.00 00	198. 00	212. 00	190. 00
Ab11	1571 .00	2025 .00	3462 .00	2171 .00	140. 00	180. 00	0.00 00	220. 00	237. 00	0.00 00	114. 00	105. 00	71.0 0
Ab12	2641 .00	1135 .00	2314 .00	3390 .00	187. 00	198. 00	0.00 00	128. 00	178. 00	0.00 00	80.0 0	149. 00	125. 00
Ab13	86.0 0	2588 .00	542. 00	5.00 00	165. 00	90.0 0	0.00 00	214. 00	132. 00	0.00 00	132. 00	186. 00	98.0 0
Ab14	40.0 0	568. 00	189. 00	7.00 00	81.0 0	119. 00	0.00 00	93.0 0	95.0 0	0.00 00	139. 00	71.0 0	313. 00
Ab15	399. 00	529. 00	970. 00	561. 00	123. 00	125. 00	0.00 00	123. 00	124. 00	0.00 00	124. 00	125. 00	123. 00
PCT-Uba(%)	55.9 8	51.0 1	55.8 8	63.0 3	1.20	1.24	0.00	1.19	1.29	0.00	1.22	1.22	1.42
COD-OU T(%)	95.8 6	89.6 3	89.5 3	94.8 8	89.9 5	89.2 1	90.4 6	92.4 3	91.5 5	91.4 5	94.2 4	89.4 7	88.9 9
TN-OUT(%)	74.8 9	69.5 0	77.0 9	78.0 1	66.5 7	81.0 9	87.4 7	63.4 7	73.0 7	74.0 7	57.7 4	42.0 2	56.0 4
TP-OUT(%)	56.0 1	42.3 0	44.3 0	48.2 9	74.2 0	66.0 6	71.4 8	81.4 0	68.5 9	60.6 5	80.2 9	72.3 2	68.1 1

*The full name and interpretation of each acronym variable are given in Table S1.

366
367

368 **Table S10**

369 Numerical results in Fig. 2a

indicators	Ba	Ab3	Ab5	PCT-Uba(%)	Ab10	Ab12	Ab15	Ab4	Ab9	Ab11	Ab13	Ab6	Ab14	COD-OUT(%)	TN-OUT(%)	PCT-Ba(%)	TP-O UT(%)
Ba	1.0	0.5	0.7	0.79	0.5	0.4	0.4	0.5	0.4	0.4	0.4	0.4	0.4	0.22	0.08	-0.45	-0.77
Ab3	0.0	1.0	0.8	0.86	0.3	0.6	0.9	0.6	0.5	0.8	0.3	0.1	0.9	0.29	-0.05	-0.82	-0.62
Ab5	0.0	0.1	1.0	0.88	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.39	0.08	-0.72	-0.66
PCT-Uba(%)	0.0	0.0	0.0	1.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.36	0.05	-0.83	-0.76
Ab10	0.0	0.0	0.0	0.83	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.46	0.29	-0.74	-0.45
Ab12	0.0	0.0	0.0	0.73	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.35	0.13	-0.53	-0.50
Ab1	0.0	0.0	0.0	0.78	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.09	0.02	-0.68	-0.65
Ab2	0.0	0.0	0.0	0.66	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-0.07	-0.23	-0.64	-0.69
Ab7	0.0	0.0	0.0	0.65	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.09	-0.13	-0.66	-0.57
Ab8	0.0	0.0	0.0	0.59	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.18	0.22	-0.60	-0.36
Ab15	0.0	0.0	0.0	0.61	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.20	-0.17	-0.69	-0.38
Ab4	0.0	0.0	0.0	0.58	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.11	0.06	-0.48	-0.61
Ab9	0.0	0.0	0.0	0.62	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.38	-0.10	-0.38	-0.47
Ab11	0.0	0.0	0.0	0.55	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-0.21	-0.33	-0.51	-0.71
Ab13	0.0	0.0	0.0	0.45	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.11	-0.41	-0.50	-0.43
Ab6	0.0	0.0	0.0	0.56	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.40	-0.01	-0.40	-0.41
Ab14	0.0	0.0	0.0	0.39	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.48	-0.04	-0.34	-0.05
COD-OUT(%)	0.0	0.0	0.0	0.36	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.00	0.37	-0.20	0.19
TN-OUT(%)	0.0	0.0	0.0	0.05	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.37	1.00	0.01	0.20
PCT-Ba(%)	0.0	0.0	0.0	-0.83	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-0.20	0.01	1.00	0.60
TP-O UT(%)	0.0	0.0	0.0	-0.76	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.19	0.20	0.60	1.00

*The full name and interpretation of each acronym variable are given in Table S1.

Table S11

Numerical results in Fig. 2b

Indicators	C/ N	TN-OUT(%)	OLR(kgC OD/m ³ d)	COD-IN (mg/L)	COD-OUT(%)	TP-IN(mg/L)	ON(%)	TP-OUT(%)	TN-IN(mg/L)	NH ₃ -IN(mg/L)
C/N	1.0 0	0.69	0.79	0.79	0.51	0.20	-0.2 5	0.02	-0.62	-0.46
TN-OUT(%)	0.6 9	1.00	0.55	0.53	0.37	0.07	-0.2 6	0.20	-0.47	-0.31
OLR(kgC OD/m ³ d)	0.7 9	0.55	1.00	1.00	0.67	0.29	-0.1 8	0.25	-0.06	0.03
COD-IN(mg/L)	0.7 9	0.53	1.00	1.00	0.66	0.33	-0.2 2	0.22	-0.07	0.04
COD-OUT (%)	0.5 1	0.37	0.67	0.66	1.00	-0.09	-0.1 9	0.19	-0.01	0.09
TP-IN(mg /L)	0.2 0	0.07	0.29	0.33	-0.09	1.00	-0.2 0	-0.06	-0.05	0.05
ON(%)	-0. 25	-0.26	-0.18	-0.22	-0.19	-0.20	1.00	0.70	0.12	-0.36
TP-OUT(%)	0.0 2	0.20	0.25	0.22	0.19	-0.06	0.70	1.00	0.18	-0.16
TN-IN(m g/L)	-0. 62	-0.47	-0.06	-0.07	-0.01	-0.05	0.12	0.18	1.00	0.88
NH ₃ -IN(mg/L)	-0. 46	-0.31	0.03	0.04	0.09	0.05	-0.3 6	-0.16	0.88	1.00

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The EMBRACE Checklist

Version 1.0. Last Updated Date: September, 2024.

The main purpose of EMBRACE Checklist is to facilitate the reporting of basic data and methods used for supervised machine learning (ML) research in environmental science and engineering (ESE). It also aims to provide a comprehensive evaluation of important preparation and analysis steps. We warmly encourage researchers to include the checklist as a reference, and if interested list it as supplementary materials in their publications.

Detailed instructions and additional resources are available open access for download from the author's GitHub: <https://github.com/starfriend10/EMBRACE>

- Future updates and new resources will be available in the above GitHub repository, which is also served as a community-owned platform to share data and codes, refine forms/checklists, discuss ML-ESE topics, promote research outcomes, and continuously identify/avoid pitfalls and follow good practices.
- We encourage you to direct share your checklists, but you can also save it as a read-only document. See instructions for detailed steps.
- Please cite the Viewpoint when using the checklist. If you have any questions or suggestions, please contact Dr. Jun-Jie Zhu at Princeton University (junjiez@princeton.edu or ranmuweijie@gmail.com).

Project Overview			
Project Title			
Contributing Authors			
Date		Completed by	
Contact Email		DOI (if published)	
Domain Category		Or Specify:	
Learning Type	Regression	Classification	Regression+ Classification
Prediction Type	Deterministic	Probabilistic	Deterministic+ Probabilistic
Other Information			

I. Problem Formulation							
Project Objective(s)							
Feasibility Assessment	Data Availability	Model Accessibility	Computation Resources	Knowledge Preparation	Time Availability	Financial Availability	Risk Tolerance
Levels							

II. Data Collection							
Data Sources		Time-series Monitoring			Field Campaigns		
		Laboratory Experiments			Simulation Outputs		
Source Types		Ordinary		Time-series		Literature-based	
Data Types		Continuous Numerical			Categorical		
		Graphical			Auditory		
Data Ethics		No special ethical considerations are required, or followed the necessary requirements when needed					
		No special permissions are required, or obtained the necessary permissions when needed.					
		Provided appropriate credit(s) to the data source(s)					
Data Availability		Data are publicly available through a PURL or DOI				Data are available as requested	
Raw Data Quantity		Raw Sample Size		Raw Variable Size		Raw Total Data volume	
Raw Data Quality		The raw data had an internal QA/QC before data retrieval?			Yes		No
		The raw data went through a peer-review or similar process?			Yes		No

III. Data Preprocessing												
Data Cleaning		Abnormal Data Management				Statistical Outlier Management				Technical Outlier Management		
		Data Correction due to Technical Limitations					Others. Specify:					
Data Enrichment		Missing Data Management			Data Augmentation			Data Aggregation			Data Balancing	
		Upper sample ratio among groups:					Others. Specify:					
Feature Engineering		Initial Variable(s) Exclusion				Feature Importance Analysis				Feature Extraction		
		Categorical Feature Transformation					Dummy Feature Removal					
		Ordinal Encoding			Encoded variable(s):							
		Feature Selection			Feature Scaling (for X)			Feature Scaling (for Y)			Circular Feature Scaling	
		New Feature Development			Ordinary Mathematical Transformation(s)				Time-Dependent Feature Transformation(s)			
				Spatial Feature Transformation(s)				Others. Specify:				
		Sparse Dataset?	Yes	No	Preprocessed to increase information density?			Yes	No			
		Normality Test (before/after transformation)					Others. Specify:					
Data Splitting	Framework			Training-Testing			(Pre-)Training-Validation-Testing				Cross-Validation-Testing	
	Splitting Method			Simple Random			Grouped Random			Simple Block		Sub-Group R/B
	Ensured training dataset covered most possible scenarios (e.g., different seasons)											
Final Data Quantity	Testing Ratio		Preprocessed Sample Size			Preprocessed Feature Size			Sample-Feature Ratio		Training SFR	

IV. Methods Selection and Model Initialization										
Methods Selection		Selected supervised learning methods to be tested based on case needs (e.g., data, computation resources, etc.)?								
	Methods Studied: N ML and M Statistical					N =		M =		
	ML Family			Tree-based			Neural Network			Others
	ML Type			Shallow ML			Deep ML		Final Optimal Method(s):	
Model Uncertainty	Examined Random Seeds				Replicated Modeling Process					
	Assessed probabilistic prediction results				Others. Specify:					

V. Model Evaluation and Model Optimization												
Model Evaluation		Set a goal model performance before evaluation										
	Specify Evaluation Metrics:											
	Studied under-fitting issue						Studied over-fitting issue					
	Final evaluation was based on the testing data						Weighted metrics were used for imbalanced data					
	ML significantly outperformed statistical models						Statistical perform similar or even better than ML models					
Hyper-parameter Optimization	HPO Method		No HPO		Trial-and-Error/Experience				Grid/Random Search (similar)			Metaheuristics
		Other HPO. Specify:										
	Number of methods tuned by HPO:					Number of hyperparameters searched:						
	Number of values in each hyperparameter:					Continuous Search			Discrete Search. Specify the number:			
	CV Method		No CV		k-fold CV			Leave-one-out CV			Stratified CV	
		Pre-defined or other CV. Specify:										

VI. Model Explanation									
Model Interpretability		Reported model interpretability as intrinsic property					Described tradeoff between interpretability and complexity		
		Explained limitations of low model interpretability					Compared the methods with different interpretabilities		
Model Explainability		Reported the explanation method(s) used for FIA			Explanation method(s):				
		Reminded high accuracy might not be trustable explanation					Described application of explanation on the optimal model		
	Reflected on understanding that explanations:				couldn't cover all mechanisms			might be incorrect	
Causality		Reflected on higher accuracy not implying logical causal relationships							
		Reported understanding that explanation results do not indicate causality							
		Described alignment of explanations with environmental domain knowledge for causality							
		Identified illogical or counterintuitive parts based on environmental domain knowledge							
		Described study of illogical or counterintuitive parts based on environmental domain knowledge							

VII. Data Leakage Management												
General/ Data Source	Verified no:			response variable Y leakage			future-to-current data leakage			current-to-current data leakage		
	Confirmed that no overlapped data between training and testing						Yes			No		
	Data splitting method for literature-source data:											
Data Enrichment	Verified that missing data replacement utilized only seen dataset						N/A		Yes		No	
	Verified that missing data interpolation utilized neighbor data in same subset						N/A		Yes		No	
	Verified that model-based imputation utilized only seen data						N/A		Yes		No	
Feature Engineering	Confirmed that feature engineering utilized only seen dataset						N/A		Yes		No	
	Confirmed that feature scaling utilized only seen dataset						N/A		Yes		No	
	Confirmed that data splitting method for time-dependent data:											
CV Loop	Verified that missing data replacement utilized only seen data in each split						N/A		Yes		No	
	Verified that missing data interpolation with neighbor data in each split						N/A		Yes		No	
	Verified that feature engineering utilized only seen data in each split						N/A		Yes		No	
	Verified that feature scaling utilized only seen data in each split						N/A		Yes		No	
	Data splitting method for literature-source data in CV loop:											
	Data splitting method for time-dependent data in CV loop:											
Others		Self-specify:										

VIII. Additional Items			
	PURL or DOI are publicly available. Data:	Code:	
	Email or URL are available to request. Data:	Code:	
	Self-specified Item:		